Simple Freight-Rate Prediction

One chronological split. One baseline. One additional model. MAE.

1. Split and clean

Train: January-June 2025, 28,806 loads. Test: July-October 2025, 19,194 loads. The later test period checks future prediction performance. No random split or cross-validation.

The 48,000 labeled rows contain 300 missing weights and 374 missing market indices. There are no duplicate load IDs or missing target prices. Weight is filled with the training median (31,400 lb). Category filling and numeric scaling also learn only from training. All observed prices ($57.22-$25,533.00) are retained; there is no evidence that extremes are errors.

2. Understand the baseline, then add one model

Baseline: predict the January-June median price, $2,022.89, for every test load. Then add Ridge: linear regression with a small fixed penalty (alpha=1), without tuning.

Method	July-October MAE
Median-price baseline	$1,138.46
One linear model (Ridge)	$143.78

MAE is the average absolute price error in dollars. Lower is better. The linear model reduces test MAE by 87.37%, so it is selected.

3. Inputs and final predictions

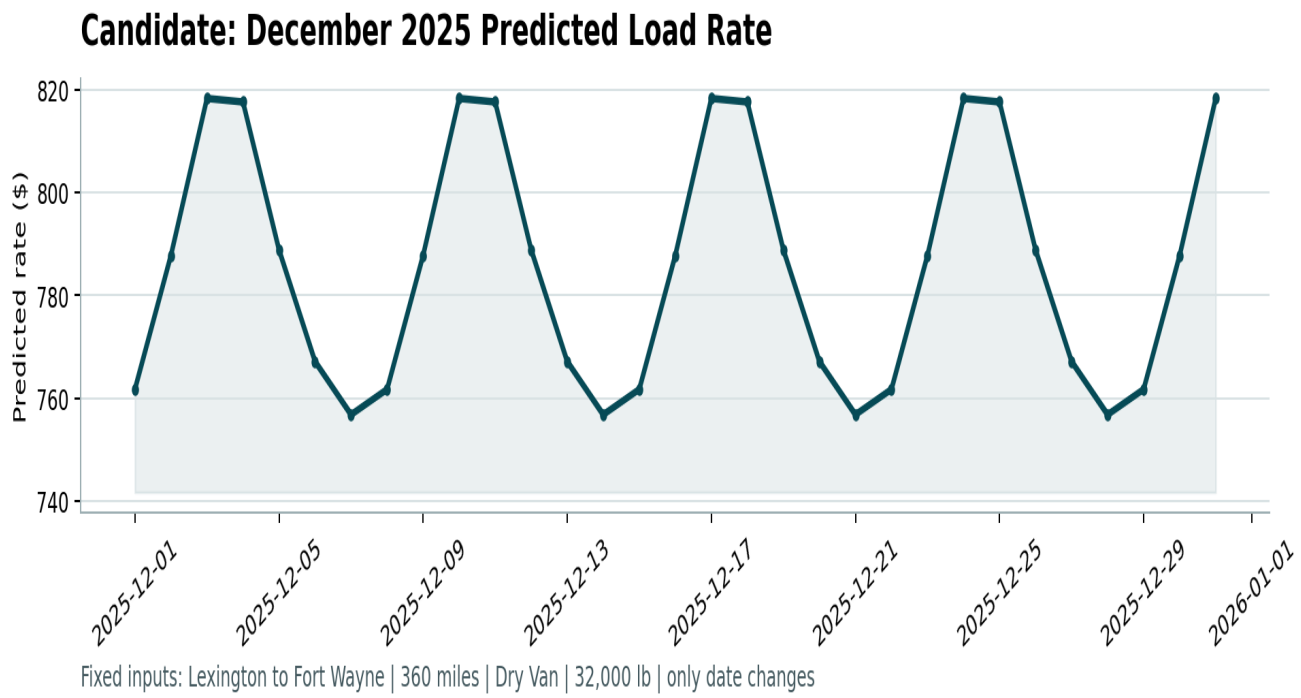
Inputs: distance, weight, pickup, delivery, equipment, and weekday from the date. Pickup, delivery, equipment, and weekday are encoded as indicator columns. Market index, quote signal, and coordinates are omitted because the December file does not supply them. Load ID only matches predictions to the template.

After evaluation, refit on all January-October labels. Preprocessing then learns from that development set only, never from November or December. This refit does not change the reported test MAE. Predictions are floored at $1 during testing and final prediction to meet the scorer requirement.

The provided scorer accepted exactly 12,000 final predictions and 31 December predictions. It checks file validity, not prediction accuracy. November targets are hidden, so November MAE is unknown.

4. Fixed December chart

This chart was generated by the supplied score.py using the completed December inputs.



Only date changes for this route. The model uses weekday, so predictions repeat weekly. It cannot capture holiday demand or seasonal changes. The July-October error is not a guarantee of December accuracy.

5. Reproduce and submit

Run python solve.py, then run the scorer command in README.md. The package includes code, dependencies, input data, both completed prediction files, measured results, the chart, and a 2-3 minute recording script. GitHub publication and the Loom recording remain submission steps.